

Property-1: The real and imaginary parts of an analytic function $w = u + iv$ satisfies the Laplace equation.

Proof. Given $f(z) = w = u + iv$ is an analytic function. Then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \rightarrow (1)$$

Diff. (1) partially w.r.t. x and y , we obtain

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial y \partial x} \rightarrow (2) \text{ and } \frac{\partial^2 v}{\partial x \partial y} = -\frac{\partial^2 u}{\partial y^2}. \rightarrow (3)$$

(2) + (3) $\Rightarrow$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial x \partial y} = 0.$$

$$\Rightarrow \nabla^2 u = 0.$$

Diff. (1) partially w.r.t. y and x , we obtain

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 v}{\partial y^2} \rightarrow (4) \text{ and } \frac{\partial^2 v}{\partial x^2} = -\frac{\partial^2 u}{\partial y \partial x}. \rightarrow (5)$$

(4) + (5) $\Rightarrow$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial x \partial y} = 0.$$

$$\Rightarrow \nabla^2 v = 0. \text{ Both } u \text{ and } v \text{ are satisfied Laplace equation.}$$

Hence proved.

Property-2: If $w = u + iv$ is an analytic function the curves of the family $u(x, y) = C_1$ and $v(x, y) = C_2$ cuts orthogonally where C_1 and C_2 are varying constants.

Proof. Given $f(z) = w = u + iv$ is an analytic function. Then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \rightarrow (1)$$

Now, $u(x, y) = C_1, \rightarrow (2)$

u is function of x and y .

Diff. (2) partially w.r.t. x and y , we obtain

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0 \Rightarrow \frac{dy}{dx} = \frac{-\partial u / \partial x}{\partial u / \partial y} = m_1(\text{say}) \rightarrow (3)$$

where m_1 is a slope of the curve.

Similarly, $v(x, y) = C_2, \rightarrow (4)$

v is function of x and y .

Diff. (4) partially w.r.t. x and y , we obtain

$$\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy = 0 \Rightarrow \frac{dy}{dx} = \frac{-\partial v / \partial x}{\partial v / \partial y} = m_2(\text{say}) \rightarrow (5)$$

where m_2 is a slope of the curve.

$$\Rightarrow m_1.m_2 = \frac{-\partial u / \partial x}{\partial u / \partial y} \cdot \frac{-\partial v / \partial x}{\partial v / \partial y} = -1. [\because \text{Using C-R equations}].$$

Therefore, the curves m_1 and m_2 are cuts orthogonally. Hence proved.

Examples: Verify that, the families of curves $u = C_1$ and $v = C_2$ cuts orthogonally when (i) z^3 (ii) $\frac{1}{z}$.

Property-3: An analytic function with constant modulus is constant.

Proof. Let $f(z) = w = u + iv$ be an analytic function.

Then, it satisfy C-R equation

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \quad \rightarrow (1)$$

$$\text{Given } |f(z)| = C(\text{constant}) \Rightarrow \sqrt{u^2 + v^2} = C \Rightarrow u^2 + v^2 = C^2. \quad \rightarrow (2)$$

Diff. (2) partially w.r.t. x and y , we get

$$2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} = 0, \quad 2u \frac{\partial u}{\partial y} + 2v \frac{\partial v}{\partial y} = 0,$$

$$\Rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} = 0, \quad u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y} = 0,$$

$$\Rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} = 0, \quad u \left(-\frac{\partial v}{\partial x} \right) + v \frac{\partial u}{\partial x} = 0,$$

The determinant of the coefficient of $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$ from the above equations is given by,

$$\begin{vmatrix} u & v \\ v & -u \end{vmatrix} = -u^2 - v^2 = -(u^2 + v^2) = C^2 \neq 0.$$

$\Rightarrow u$ and v are independent of x and y .

$$\text{Then, } \frac{\partial u}{\partial x} = 0 \text{ and } \frac{\partial v}{\partial x} = 0 \Rightarrow \frac{\partial v}{\partial y} = 0 \text{ and } \frac{\partial u}{\partial y} = 0.$$

Therefore, u and v are constant which gives $f(z) = u + iv$ is constant.

Hence proved.

Property-4: Show that the analytic function with constant real part is constant.

Proof. Let $f(z) = w = u + iv$ be an analytic function.

Then, it satisfy C-R equation

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \quad \rightarrow (1)$$

Given $u = C_1(\text{constant}) \quad \rightarrow (2)$

Diff. (2) partially w.r.t. x and y , we get

$$\frac{\partial u}{\partial x} = 0 \text{ and } \frac{\partial u}{\partial y} = 0.$$

By the equation (1), we obtain

$$\frac{\partial v}{\partial y} = 0 \text{ and } \frac{\partial v}{\partial x} = 0.$$

$\Rightarrow v$ is independent of x and y .

So, v is also constant. That is, $v = C_2(\text{constant})$.

Then, $f(z) = u + iv = C_1 + iC_2$ is constant.

Hence proved.

Property-5: Show that the analytic function with constant imaginary part is constant.

Property-6: If $f(z)$ and $f(\bar{z})$ are analytic then $f(z)$ is constant.

Some problems based on properties.

If $f(z)$ is an analytic function of z , prove that

- ① $\nabla^2(\log|f(z)|) = 0, f'(z) \neq 0,$
- ② $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) [Re(f(z))]^2 = 2|f'(z)|^2,$
 $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) [Im(f(z))]^2 = 2|f'(z)|^2,$
- ③ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4|f'(z)|^2,$
- ④ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^p = p^2 |f(z)|^{p-2} |f'(z)|^2,$
- ⑤ $\left(\frac{\partial|f(z)|}{\partial x}\right)^2 + \left(\frac{\partial|f(z)|}{\partial y}\right)^2 = |f'(z)|^2.$

An additional proof

. If $z = x + iy$ and $\bar{z} = x - iy$ then $Re(z) = x = \frac{z+\bar{z}}{2}$ and $Im(z) = y = \frac{z-\bar{z}}{2i}$.

Also, $\frac{\partial x}{\partial z} = \frac{1}{2}$ and $\frac{\partial y}{\partial z} = \frac{1}{2i}$.

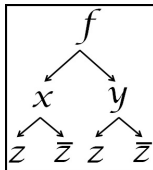
Now,

$$\begin{aligned}\frac{\partial f}{\partial z} &= \frac{\partial f}{\partial x} \left(\frac{\partial x}{\partial z} \right) + \frac{\partial f}{\partial y} \left(\frac{\partial y}{\partial z} \right) = \frac{\partial f}{\partial x} \left(\frac{1}{2} \right) + \frac{\partial f}{\partial y} \left(\frac{1}{2i} \right) \\ &= \frac{\partial f}{\partial x} \left(\frac{1}{2} \right) + \frac{\partial f}{\partial y} \left(\frac{1}{2i} \times \frac{i}{i} \right) = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) \rightarrow (1)\end{aligned}$$

$$\begin{aligned}\frac{\partial f}{\partial \bar{z}} &= \frac{\partial f}{\partial x} \left(\frac{\partial x}{\partial \bar{z}} \right) + \frac{\partial f}{\partial y} \left(\frac{\partial y}{\partial \bar{z}} \right) = \frac{\partial f}{\partial x} \left(\frac{1}{2} \right) + \frac{\partial f}{\partial y} \left(\frac{-1}{2i} \right) \\ &= \frac{\partial f}{\partial x} \left(\frac{1}{2} \right) + \frac{\partial f}{\partial y} \left(\frac{-1}{2i} \times \frac{i}{i} \right) = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \rightarrow (2)\end{aligned}$$

$$\begin{aligned}\therefore \frac{\partial^2 f}{\partial z \partial \bar{z}} &= \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) \times \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \\ &= \frac{1}{4} \left(\frac{\partial^2 f}{\partial x^2} + i \frac{\partial^2 f}{\partial x \partial y} - i \frac{\partial^2 f}{\partial y \partial x} + \frac{\partial^2 f}{\partial y^2} \right) = \frac{1}{4} \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right)\end{aligned}$$

$$\Rightarrow \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) = \nabla^2 f = 4 \frac{\partial^2 f}{\partial z \partial \bar{z}}.$$



1. Prove that $\nabla^2(\log|f(z)|) = 0$, $f'(z) \neq 0$.

Proof. WKT $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$.

$$\begin{aligned} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (\log(|f(z)|)) &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} (\log(|f(z)|)) \\ &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} \frac{1}{2} (\log(|f(z)|^2)) \\ &= \frac{4}{2} \frac{\partial^2}{\partial z \partial \bar{z}} (\log(f(z)f(\bar{z}))) \\ &= 2 \frac{\partial^2}{\partial z \partial \bar{z}} (\log(f(z)) + \log(f(\bar{z}))) \\ &= 2 \frac{\partial}{\partial z} \left((0) + \frac{1}{f(\bar{z})} f'(\bar{z}) \right) = 2(0 + 0) = 0 \end{aligned}$$

Hence, $\nabla^2(\log(|f(z)|)) = 0$.

2. (i) Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) [Re(f(z))]^2 = 2|f'(z)|^2$.

Proof. WKT $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$.

$$\begin{aligned} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) [Re(f(z))]^2 &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} [Re(f(z))]^2 \\ &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} \left[\frac{f(z) + f(\bar{z})}{2} \right]^2 \\ &= 4 \times \frac{1}{4} \frac{\partial^2}{\partial z \partial \bar{z}} [f(z) + f(\bar{z})]^2 \\ &= \frac{\partial}{\partial z} 2(f(z) + f(\bar{z})) f'(\bar{z}) \\ &= 2f'(z)f'(\bar{z}) = 2|f'(z)|^2. \end{aligned}$$

Hence, $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} [Re(f(z))]^2 = 2|f'(z)|^2$.

2. (ii) Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) [Im(f(z))]^2 = 2|f'(z)|^2$.

Proof. WKT $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\frac{\partial^2}{\partial z\partial\bar{z}}$.

$$\begin{aligned} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) [Im(f(z))]^2 &= 4\frac{\partial^2}{\partial z\partial\bar{z}} [Im(f(z))]^2 \\ &= 4\frac{\partial^2}{\partial z\partial\bar{z}} \left[\frac{f(z)-f(\bar{z})}{2i}\right]^2 \\ &= 4 \times \frac{1}{4(i)^2} \frac{\partial^2}{\partial z\partial\bar{z}} [f(z) - f(\bar{z})]^2 \\ &= -1 \times \frac{\partial}{\partial z} (-2(f(z) - f(\bar{z}))) f'(\bar{z}) \\ &= 2f'(z)f'(\bar{z}) = 2|f'(z)|^2. \end{aligned}$$

Hence, $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} [Im(f(z))]^2 = 2|f'(z)|^2$.

3. Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4|f'(z)|^2$,

Proof. WKT $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\frac{\partial^2}{\partial z\partial\bar{z}}$.

$$\begin{aligned} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) (|f(z)|^2) &= 4\frac{\partial^2}{\partial z\partial\bar{z}} (|f(z)|^2) \\ &= 4\frac{\partial^2}{\partial z\partial\bar{z}} (f(z)f(\bar{z})) \\ &= 4\frac{\partial}{\partial\bar{z}} (f(z)f'(\bar{z})) \\ &= 4(f'(z)f'(\bar{z})) = 4|f'(z)|^2 \end{aligned}$$

Hence, $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4|f'(z)|^2$.

4. Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^p = p^2 |f(z)|^{p-2} |f'(z)|^2$,

Proof. WKT $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$.

$$\begin{aligned}
 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) (|f(z)|^p) &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} (|f(z)|^p) \\
 &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} (|f(z)|^2)^{\frac{p}{2}} \\
 &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} (f(z) f(\bar{z}))^{\frac{p}{2}} \\
 &= 4 \frac{\partial^2}{\partial z \partial \bar{z}} (f(z))^{\frac{p}{2}} (f(\bar{z}))^{\frac{p}{2}} \\
 &= 4 \frac{\partial}{\partial z} ((f(z))^{\frac{p}{2}}) \frac{\partial}{\partial \bar{z}} ((f(\bar{z}))^{\frac{p}{2}}) \\
 &= 4 \left(p/2 (f(z))^{p/2-1} f'(z)\right) \left(\frac{p}{2} (f(\bar{z}))^{\frac{p}{2}-1} f'(\bar{z})\right) \\
 &= p^2 (f'(z) f'(\bar{z})) \left((f(z) f(\bar{z}))^{\frac{p}{2}-1}\right) \\
 &= p^2 |f'(z)|^2 (|f(z)|^2)^{\frac{p-2}{2}}
 \end{aligned}$$

Hence, $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) (|f(z)|^p) = p^2 |f(z)|^{p-2} |f'(z)|^2$.

5. Prove that $\left(\frac{\partial|f(z)|}{\partial x^2}\right)^2 + \left(\frac{\partial|f(z)|}{\partial y^2}\right)^2 = |f'(z)|^2$,

Proof. Let $f(z) = u + iv$, $|f(z)| = \sqrt{u^2 + v^2}$.

Also, $f'(z) = u_x + iv_x \Rightarrow |f'(z)|^2 = u_x^2 + v_x^2$ and $f'(z) = u_y + iv_y \Rightarrow |f'(z)|^2 = u_y^2 + v_y^2$

Now, $\frac{\partial}{\partial x}|f(z)| = \frac{1}{2\sqrt{u^2+v^2}} (2uu_x + 2vv_x) = \frac{(uu_x + vv_x)}{\sqrt{u^2+v^2}}$

$$\Rightarrow \left(\frac{\partial}{\partial x}|f(z)|\right)^2 = \frac{(uu_x + vv_x)^2}{(u^2 + v^2)} \rightarrow (1).$$

Similarly, $\left(\frac{\partial}{\partial y}|f(z)|\right)^2 = \frac{(uu_y + vv_y)^2}{(u^2 + v^2)} \rightarrow (2).$

$$\begin{aligned} (1) + (2) &\Rightarrow \left(\frac{\partial}{\partial x}|f(z)|\right)^2 + \left(\frac{\partial}{\partial y}|f(z)|\right)^2 = \frac{(uu_x + vv_x)^2 + (uu_y + vv_y)^2}{(u^2 + v^2)} \\ &= \frac{(u^2 u_x^2 + v^2 v_x^2 + 2uu_x vv_x) + (u^2 u_y^2 + v^2 v_y^2 + 2uu_y vv_y)}{(u^2 + v^2)} \\ &= \frac{u^2(u_x^2 + u_y^2) + v^2(v_x^2 + v_y^2) + 2uv(u_x v_x + u_y v_y)}{(u^2 + v^2)} \\ &= \frac{u^2(u_x^2 + (-v_x)^2) + v^2((-u_y)^2 + v_y^2) + 2uv(-u_x u_y + u_y u_x)}{(u^2 + v^2)} \quad [\because \text{C-R equations}] \\ &= \frac{u^2(u_x^2 + v_x^2) + v^2(u_y^2 + v_y^2) + 2uv(0)}{(u^2 + v^2)} \\ &= \frac{u^2|f'(z)|^2 + v^2|f'(z)|^2}{(u^2 + v^2)} \\ &= \frac{u^2|f'(z)|^2 + v^2|f'(z)|^2}{(u^2 + v^2)} \\ &= \frac{(u^2 + v^2)}{(u^2 + v^2)} |f'(z)|^2 = |f'(z)|^2. \end{aligned}$$

Hence, $\left(\frac{\partial|f(z)|}{\partial x^2}\right)^2 + \left(\frac{\partial|f(z)|}{\partial y^2}\right)^2 = |f'(z)|^2$.